Steps	Reason
7. $W(P) \Rightarrow O(P)$	Universal instantiation on 2
8. $O(P)$	Modus ponens on 6,7
9. $C(P) \land O(P)$	Conjunction on 5,8
10. $\exists x(C(x) \land O(x))$	Existential generalization on 9

c) "Each of the 93 students in this class owns a personal computer. Everyone who owns a personal computer can use a word processing program. Therefore Zeke, a student in this class, can use a word processing program.

$\boxed{\text{Soln:}}$ $S(x)$: x is a student in this class $PC(x)$: x owns a personal computer
$WP(x)$: x can use a word processing program.

$\boxed{\text{Domain:}}$ Set of all people in the world

$\boxed{\text{Premise:}}$ ① $\forall x(S(x) \Rightarrow PC(x))$ ② $\forall x(PC(x) \Rightarrow WP(x))$ ③ $S(Zeke)$

$\boxed{\text{Conclusion:}}$ $S(Zeke) \land WP(Zeke)$

Steps	Reason
1. $\forall x(S(x) \Rightarrow PC(x))$	Premise
2. $\forall x(PC(x) \Rightarrow WP(x))$	Premise
3. $S(Zeke)$	Premise
4. $S(Zeke) \Rightarrow PC(Zeke)$	Universal instantiation on 1
5. $PC(Zeke) \Rightarrow WP(Zeke)$	Universal instantiation on 2
6. $PC(Zeke)$	Modus ponens on 3,4
7. $WP(Zeke)$	Modus ponens on 6,5
8. $S(Zeke) \land WP(Zeke)$	Conjunction on 3,7

d) "Everyone in New Jersey lives within 50 miles of the ocean. Someone in New Jersey has never seen the ocean. Therefore, someone who lives within 50 miles of the ocean has never seen the ocean."